

ICLR 2027 — Thorough Review & Technical Audit

ONE TOKEN OWNS THE CALIBRATION HESSIAN: WHY LAYER-WISE QUANTIZATION SILENTLY BREAKS
INSTRUCTION FOLLOWING

Strict reviewer perspective — Logic, Grammar, Citations, Numbers, Coherence

Recommendation

7.5 – 8 (Accept / Clear Accept)

1. Paper Summary

The paper identifies a silent failure of GPTQ at 3 bits under the short-document calibration protocol used by default in GPTQModel and LLM Compressor. Instruction following collapses (Llama-3.1-8B-Instruct IFEval 0.768 → 0.155; Qwen2.5-14B-Instruct 0.820 → 0.426) while MMLU stays competitive and perplexity remains ordinary. The damage is localized to the MLP down-projection(s) of the attention-sink layer, where the calibration Hessian is rank-1 and owned by BOS, while the loss depends on template tokens. An exact closed-form under a dominant token (Theorem 1) predicts the lesion; transplants confirm it; a simple Hessian reweighting (output sensitivity / input energy) repairs every 3-bit collapse found.

2. Logic & Causal Coherence

What holds tightly

- Observation → localization by transplants → closed-form under dominant token → verified signatures → derived repair is a clean causal chain for *lesion formation*.
- Theorem 1 is exact under the stated assumption ($\kappa \blacksquare 1$). Direction, row concentration and amplitude overshoot are confirmed across the census.
- Pre-registration of post-diagnosis experiments + explicit list of 30 failed predictions is excellent scientific practice and increases trust in the surviving claims.

Material logical gaps

1. **Absorption is unexplained.** The paper repeatedly states that the lesion is necessary but not sufficient; collapse occurs only when later matrices fail to absorb it. This absorption process is measured (transplants) but never derived or even modelled. For a paper whose title claims to explain *why* instruction following breaks, this is the largest open causal hole. Section 4 and the limitations section acknowledge it, but the main narrative still presents the story as essentially complete.
2. **Readout prediction only partially holds.** Corollary 1(ii) places template tokens well on Qwen (recall 0.52) but poorly on Llama (recall 0.11). One of the three core signatures is only half-confirmed. This weakens the claim that the closed-form fully accounts for the observed damage pattern.
3. **Mistral phenotype is incompletely resolved.** Collapse requires the full set of 14 matrices of layers 0–1; neither down-projection alone is sufficient. The single-matrix (or pair-of-down-projections) story does not cover this case. The paper localizes but does not explain it.
4. **Loop correction (Appendix B.2) changed key results.** An earlier defective implementation produced different numbers (Llama under Pile windows collapsed; foreign lesion was tolerated). After correction the protocol dependence and transplant conclusions changed. The authors are honest, but the fact that a bug altered the scientific conclusions should be more visible in the main text, not buried in the appendix.

3. Grammar, Style & Clarity

The writing is dense and information-packed. Many sentences are extremely long (especially in the Introduction, Section 4 and Section 5). This is not fatal, but it will cost readability for some reviewers. Specific issues:

- **Sentence length.** Example (Intro): “Three properties explain why this has not been reported: the failure is silent, because the metrics never present the tokens that carry the damage; it is rare, two of the 18 instruction-tuned models we tested under the default protocol; and it depends on how the calibration text is cut...” — better as three shorter sentences.
- **Compressed lists.** Phrases such as “the failure is rare, two of eighteen models under that protocol” omit the verb and feel telegraphic.
- **Article / determiner drops.** Occasional missing articles (“under dominant token”, “on Qwen it alone collapses”) that arise from aggressive compression.
- **Abstract vs body precision.** Abstract rounds 0.768→0.77, 0.155→0.15, 0.820→0.82, 0.426→0.43. Minor, but unnecessary in a precision-critical paper.

Overall grammar is correct; the problems are stylistic density and a few telegraphic constructions, not grammatical errors.

4. Citations & Related Work

What is properly cited

- Core GPTQ / OBS lineage: Frantar & Alistarh 2022, Frantar et al. 2023, LeCun et al. 1990, Hassibi & Stork 1993 — present and correctly used.
- Attention sinks: Xiao et al. 2024, Sun et al. 2024, Gu et al. 2025, Yu et al. 2024 (super weights), Chen et al. 2026b/c — present.
- Calibration sensitivity, chat-formatted calibration, length normalization, surveys on GPTQ vs RTN, and papers showing likelihood metrics miss task damage — all cited.
- Competing quantizers (AWQ, HQQ, QuaRot, AutoRound, SpQR, etc.) are discussed and compared.

Issues / missing pieces

1. **Critical TODO left in the paper.** Reproducibility Statement ends with “[TODO: Anonymous repository link.]”. This must be fixed before camera-ready (and ideally before rebuttal). A missing code link is a common reason for lowered scores.
2. **Future-dated citations.** Many references are 2025–2026 arXiv preprints. This is consistent with an ICLR 2027 submission timeline, but reviewers may flag any that are not yet public or peer-reviewed. No action required if they are real; just be prepared to defend them.
3. **No obvious foundational omissions** for the core claims. The related-work coverage of both quantization and attention-sink literature is solid.

5. Number Consistency Audit

All major numbers were cross-checked across abstract, introduction, Tables 3–5, 7–11, 15 and the body. No contradictions found. Only cosmetic rounding appears in the abstract.

Quantity	Abstract	Body / Tables	Verdict
Llama IFEval	0.77 → 0.15	0.768 → 0.155 (RTN 0.565)	OK (rounding)
Qwen IFEval	0.82 → 0.43	0.820 → 0.426 (RTN 0.697)	OK (rounding)
Re-round lesion L/Q	—	0.622 / 0.768	Consistent
Corrected objective	—	0.638 / 0.736	Consistent
Window recovery	0.645 (Llama)	0.645 / 0.766	Consistent
Point drop vs RTN	—	41 pts / 27 pts	Arithmetic OK

Census Table 15 is internally consistent with the main claims. Multi-seed Table 16 also matches the text discussion of the exceptional healthy Llama draw (0.531).

6. Other Technical & Presentation Issues

- **Simulated quantization only.** All results use fake-quant on the integer grid kept in 16-bit storage. Real integer kernels, packing, and kernel-specific numerics can change behaviour, especially at 3 bits. This should be stated more prominently (currently mostly in Appendix B).
- **Repair cost.** On Llama the corrected objective recovers IFEval but only partially recovers GSM8K (0.548 vs 0.632 for re-rounding) and slightly hurts MMLU relative to the collapsed checkpoint. Presenting the method as a clean solution slightly overstates the practical win.
- **Figure / Table references.** All major claims point to the correct figures and tables; no broken references observed.
- **Double-blind compliance.** Authors are anonymous; repository link is still a TODO. Otherwise clean.

7. Final Recommendation & Revision List

Accept (7.5–8). High-quality mechanistic work on a practically important, previously unreported failure mode. The localization experiments and the closed-form under a dominant token are strong. The incomplete absorption story, the partial readout failure, the unresolved Mistral case, and the non-zero cost of the repair are real limitations, but the honesty about failed predictions and the pre-registration raise the paper.

Must-fix before camera-ready / strong revision requests

1. Replace “[TODO: Anonymous repository link.]” with the actual anonymized link (or a clear statement that code will be released upon acceptance).

2. Soften abstract numbers to match table precision, or explicitly note that they are rounded.
3. Move a short “limitations of the theory” paragraph (absorption + readout + Mistral) earlier — ideally into Section 5 or the end of Section 4 — so the reader does not reach the end believing the causal account is complete.
4. Mention the loop-correction episode more visibly (one sentence in the main experimental section or in the caption of the relevant figure/table).
5. Flag more prominently that all quantization is simulated (fake-quant).
6. Break the longest sentences in the Introduction and Section 4; the density currently hurts readability.
7. Optional but useful: a one-paragraph practical recommendation box for practitioners (when to use Eq. 8 / chat calibration / simply re-round the sink matrix).